

Statistical Learning

Final Project Topics and Expectations

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Project expectations

Projects may be undertaken individually or in groups of two. Each project should develop a topic sufficiently carefully that other students in the course can learn it from your lectures and report. These are some guidelines for the final presentation and project report.

- **Class lectures.** An individual project should be presented in one class lecture. A two-person group should give two class lectures, with each member responsible for one lecture. The lectures should form a coherent sequence covering the problem, the Bayesian formulation, the main computational ideas, and illustrative examples or experiments.
- **Lecture-note-style report.** Submit a self-contained report written as lecture notes for students in this course. Introduce the problem and notation, explain the prior and likelihood, derive the important formulas, describe the algorithm, and discuss examples, results, and limitations. Include references and clearly distinguish existing methods from your own implementation, comparisons, or extensions.
- **Computation.** Implement a representative method and include reproducible code for your experiments. Explain the central inference algorithm; existing routines may be used for standard numerical operations and specialized sampling steps. Comparisons should account for tuning and computational budgets.

The emphasis is on understanding, clear exposition, and a thoughtful computational investigation.

How to approach a project

Each project should be organized around a concrete statistical or computational problem rather than around an algorithm alone. Begin by explaining the problem and the standard approach to it. Then explain what changes when the problem is formulated probabilistically or Bayesianly: what is treated as unknown, what information is observed, what prior or structural assumptions are introduced, and what posterior or algorithm results.

An important part of the project is to identify what the probabilistic formulation provides that the standard approach does not. Depending on the problem, this may include uncertainty quantification, robustness, adaptation, a principled rule for choosing future computations, or the ability to stop a numerical algorithm before it has run to completion.

The computational investigation should then identify the regimes in which these advantages are genuine and the regimes in which the additional modelling assumptions or computational cost become disadvantages. Thus the goal is not simply to implement a Bayesian method, but to understand *why* one might use it and what determines whether it works well.

Main references

- Philipp Hennig, Michael A. Osborne, and Hans P. Kersting (2022), *Probabilistic Numerics: Computation as Machine Learning*. Cambridge University Press. Part I supplies background on Gaussian models, regression, filtering, and hierarchical inference for Projects 1–5.
- Alice Cima (2024), *Nonparametric Regression with Cauchy Priors*. Ph.D. dissertation, University of California, Berkeley.

1 Bayesian Numerical Integration

Suggested reading: Probabilistic Numerics, Part II (Chapters 8–13).

Central problem. Suppose we want to compute

$$F = \int f(x) d\nu(x),$$

but evaluating f is expensive, so that we can observe it at only a limited number of locations $x_1, \dots, x_n$. The fundamental problem is how to use these finitely many function evaluations to estimate the integral, and how to quantify the remaining numerical error.

A classical quadrature rule produces an approximation of the form

$$\hat{F} = \sum_{i=1}^n w_i f(x_i),$$

with nodes and weights chosen to have good approximation properties. Monte Carlo integration instead chooses evaluation points randomly and averages the resulting function values. These methods are usually studied through convergence rates or deterministic or probabilistic error bounds.

Probabilistic approach. Bayesian quadrature treats the unknown integrand itself as an object about which we have uncertainty. For example, place a Gaussian process prior

$$f \sim GP(m, k).$$

After observing $f(x_1), \dots, f(x_n)$, the posterior distribution on f induces a posterior distribution on the integral F . Its posterior mean gives a numerical approximation, while its posterior variance quantifies the uncertainty remaining after the available function evaluations.

The important distinction is therefore not simply “Bayesian versus classical quadrature.” Certain classical quadrature rules can themselves be recovered from particular probabilistic models. The broader question is whether prior knowledge about the structure of the integrand can be used to extract more information from each expensive evaluation.

Key questions to study. How much is gained by using smoothness or other structural information about f ? How sensitive are the estimate and its uncertainty to the kernel? What happens when the assumed smoothness is wrong? How should evaluation locations be chosen, and can the posterior uncertainty guide the choice of the next location?

Suggested tasks.

- Derive the posterior mean and variance of the integral from function evaluations.
- Implement Bayesian quadrature and compare it with a classical quadrature rule and Monte Carlo integration under equal function-evaluation budgets.
- Study the effects of the kernel, smoothness assumptions, and evaluation locations on both numerical error and posterior uncertainty.
- Include examples for which the Gaussian process prior is well matched to the integrand and examples that deliberately violate its smoothness assumptions.
- Examine whether posterior credible intervals for the integral are well calibrated relative to the actual numerical error.
- Optionally, explain how a familiar classical quadrature rule arises from a particular prior, or study sequential selection of evaluation points.

2 Probabilistic Numerical Linear Algebra

Suggested reading: Probabilistic Numerics, Part III (Chapters 14–23).

Central problem. Consider a large symmetric positive-definite linear system

$$Ax = b.$$

For moderate problems one can use a direct factorization, but for very large matrices this may be too expensive. Iterative methods such as conjugate gradients instead produce a sequence

$$x_0, x_1, x_2, \dots$$

of increasingly accurate approximations using matrix–vector products.

The classical question is how quickly x_k approaches the exact solution $x = A^{-1}b$. A related practical problem is deciding when the current approximation is accurate enough to stop the computation.

Probabilistic approach. A probabilistic linear solver interprets computations such as

$$y_k = As_k$$

as observations about an unknown object. Depending on the formulation, the unknown may be A , its inverse A^{-1} , or the solution x itself. A prior is updated sequentially as matrix–vector products are computed.

For suitable choices of prior and search directions, the resulting posterior mean is closely connected to classical projection methods and to conjugate gradients. Thus the aim is not necessarily to replace conjugate gradients by a completely different solver. Rather, one asks whether the same

computation can also provide meaningful information about the error that remains after only k iterations.

Key questions to study. What exactly should be treated as unknown: A , A^{-1} , or x ? What information does each matrix–vector product reveal? How does the condition number or spectrum of A affect convergence? Can posterior uncertainty reliably describe the remaining solution error when the algorithm is stopped early?

This last issue is particularly important. An $N \times N$ matrix contains N^2 entries, whereas a modest number of matrix–vector products reveals only limited information about the matrix. Obtaining calibrated uncertainty can therefore be considerably more difficult than obtaining a good point estimate.

Suggested tasks.

- Formulate $Ax = b$ as an inference problem. Clearly identify the uncertain object, its prior, and the information supplied by a matrix–vector product.
- Derive the posterior or sequential update for a simple Gaussian formulation.
- Implement a probabilistic solver for symmetric positive-definite systems and explain its relation to conjugate gradients.
- Compare the probabilistic posterior mean with the classical iterative estimate under the same matrix–vector-product budget.
- Study matrices with different spectra and condition numbers.
- Stop the algorithm after different numbers of iterations and compare the true error

$$\|x_k - x\|$$

with the posterior uncertainty reported by the probabilistic solver.

- Investigate the effect of the prior or preconditioner on both convergence and uncertainty calibration.

3 Probabilistic Local Optimization

Suggested reading: Probabilistic Numerics, Part IV (Chapters 24–28).

Central problem. Suppose we want to minimize a loss function

$$L(w).$$

Given a current iterate w and a search direction d , an optimization algorithm must decide how far to move:

$$w_{\text{new}} = w + \alpha d.$$

In deterministic optimization, a line search evaluates the loss and its derivative along this direction and selects a step size satisfying conditions such as the Wolfe conditions.

In modern statistical learning, however, losses and gradients are often estimated from minibatches. Their evaluations are therefore noisy. A classical line search may reject a good step, or accept a bad one, simply because the observed function value or directional derivative contains sampling noise.

Probabilistic approach. A probabilistic line search treats the unknown one-dimensional function

$$\phi(\alpha) = L(w + \alpha d)$$

and possibly its derivative as uncertain. Noisy evaluations of the loss and gradient update a probabilistic model for ϕ . Candidate step sizes can then be assessed according to the probability that appropriate acceptance conditions hold.

The essential difference from an ordinary deterministic line search is that the algorithm does not treat observed minibatch losses and gradients as exact. Their uncertainty enters directly into the decision of whether to accept a step and where to evaluate next.

Key questions to study. When is explicitly modelling evaluation noise useful? How does performance depend on minibatch size and noise level? How much additional computation is required to maintain the probabilistic model? Does the method reduce the number of expensive loss or gradient evaluations required to reach a given optimization accuracy?

Suggested tasks.

- Explain the probabilistic line-search formulation and the model placed on $\phi(\alpha)$ and $\phi'(\alpha)$.
- Explain how uncertainty enters the probability of satisfying the line-search acceptance conditions.
- Implement a simplified probabilistic line search for a small statistical learning problem.
- Compare with a fixed learning rate, a scheduled learning rate, and a conventional deterministic line search.
- Vary the minibatch size or noise level systematically.
- Measure optimization performance both as a function of iteration number and of function/gradient evaluation cost.
- Investigate the regime in which the probabilistic line search begins to provide an advantage over a conventional method.

4 Bayesian Global Optimization

Suggested reading: Probabilistic Numerics, Part V (Chapters 29–34).

Central problem. Suppose we wish to minimize an objective function

$$f(x),$$

but evaluating f is expensive. A single evaluation might require fitting a statistical model, running a large simulation, or performing a physical experiment. The limiting resource is therefore the number of evaluations of f .

Grid search evaluates a predetermined collection of points. Random search samples candidate points independently. Neither method makes full use of the information contained in previous evaluations when deciding where to evaluate next.

Probabilistic approach. Bayesian optimization places a probabilistic surrogate model, commonly

$$f \sim GP(m, k),$$

on the unknown objective. After each evaluation, the posterior tells us both where the objective appears promising and where uncertainty remains large.

An acquisition function converts this posterior into a decision about the next evaluation. The central tension is between

exploitation: evaluating where f currently looks good

and

exploration: evaluating where uncertainty is large.

Expected improvement, probability of improvement, and confidence-bound methods encode different versions of this tradeoff.

The probabilistic model therefore plays two roles: it provides an estimate of the objective between observed locations, and it determines which expensive computation should be performed next.

Key questions to study. How much can adaptive evaluation improve upon random or grid search when the evaluation budget is small? How sensitive is performance to the kernel, length scale, initial design, and acquisition function? What happens when the objective violates the smoothness assumptions of the Gaussian process? How quickly does performance deteriorate as dimension increases?

Suggested tasks.

- Implement Gaussian-process-based Bayesian optimization.
- Explain and compare two acquisition functions, such as expected improvement and a confidence-bound rule.
- Compare Bayesian optimization with random search and grid search under the same evaluation budget.
- Plot the best objective value found as a function of the number of function evaluations.
- Study sensitivity to the initial design and Gaussian process hyperparameters.
- Compare a smooth test function well matched to the GP model with an example for which the model is deliberately misspecified.
- Optionally, apply the method to tuning one or two hyperparameters of a statistical learning algorithm.

5 Probabilistic ODE Solving

Suggested reading: Probabilistic Numerics, Part VI (Chapters 35–41).

Central problem. Consider an initial-value problem

$$y'(t) = f(t, y(t)), \quad y(0) = y_0.$$

Except for special differential equations, the exact solution curve is not available. Classical methods such as Euler’s method and Runge–Kutta methods approximate the solution on a discrete time grid. Smaller step sizes usually produce greater accuracy, but require more evaluations of the vector field.

A classical numerical solver therefore produces an approximate trajectory, together with theoretical results concerning its local or global discretization error.

Probabilistic approach. A probabilistic ODE solver treats the unknown deterministic solution curve as a latent function. A stochastic-process prior is placed jointly on the solution and its derivatives. The differential equation itself supplies information through the relation

$$y'(t) = f(t, y(t)).$$

Sequential filtering can then be used to update the distribution over the solution as the solver moves through time.

The resulting method produces both a numerical approximation and a distribution representing the uncertainty due to incomplete computation. This uncertainty should not be confused with randomness in the physical system: the true ODE solution is deterministic. The uncertainty describes our incomplete knowledge of that solution after finitely many numerical evaluations.

Key questions to study. How does uncertainty behave as the step size changes? Does the posterior become appropriately wider when the numerical approximation becomes less accurate? How does the order or smoothness of the prior affect performance? What approximation is required when f is nonlinear? How do stability and long-time error accumulation affect uncertainty calibration?

Suggested tasks.

- Explain the stochastic-process prior on the solution and its derivatives.
- Explain precisely how the differential equation provides observations or constraints for posterior updating.
- Implement a basic Gaussian ODE filter and identify the approximation used for nonlinear differential equations.
- Compare with Euler and Runge–Kutta methods on equations having known or highly accurate reference solutions.
- Vary the step size and study both numerical error and posterior uncertainty.
- Investigate whether posterior intervals contain the reference solution at roughly the expected frequency.
- Smoothing or uncertainty-based adaptive step-size selection can be pursued as an optional extension.

6 Cauchy Difference Priors: Smoothing and Learning the Global Scale

Suggested reading: Cima’s thesis, Chapter 1, especially Section 1.3 and Appendix A.1.

Central problem. Consider the Gaussian sequence model

$$Y_i = \theta_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

We want to estimate an underlying signal $\theta = (\theta_1, \dots, \theta_n)$ that may contain long smooth or nearly constant regions together with a small number of abrupt changes.

A Gaussian difference prior strongly penalizes large differences and is therefore effective for smooth signals, but can oversmooth genuine jumps. Methods such as total variation denoising allow sharp changes more readily, but produce a point estimate and require the amount of regularization to be selected.

Bayesian approach. Place a heavy-tailed prior on discrete differences:

$$(D^{(k)}\theta)_j \stackrel{\text{iid}}{\sim} \text{Cauchy}(0, \tau),$$

where $D^{(k)}$ is the k th difference operator.

The sharp peak of the Cauchy distribution around zero encourages most differences to be small, while its heavy tails allow a small number of differences to be large without excessive shrinkage. The model therefore attempts to combine aggressive noise reduction with preservation of genuine jumps.

The global scale τ controls the overall amount of regularization. Small τ favors stronger smoothing, while larger τ permits more variation. A central issue is therefore whether τ should be fixed, estimated by empirical Bayes, or assigned a prior and learned jointly with the signal.

Key questions to study. Can the heavy-tailed prior adapt simultaneously to flat, smoothly varying, and abruptly changing regions? How sensitive is the estimate to τ ? Does learning τ improve performance over fixing it? How well calibrated are posterior credible intervals? How quickly does the Gibbs sampler mix, and how does computation scale with n ?

Suggested tasks.

- Explain the Gaussian scale-mixture representation of the Cauchy prior and derive the corresponding Gibbs sampler.
- Begin with $k = 1$, known σ , and proper priors on the initial level and global scale.
- Compare several fixed choices of τ with fully Bayesian inference for τ .
- Investigate empirical Bayes estimation of τ by marginal likelihood as an extension, for example through Monte Carlo EM.
- Compare with TV denoising and a Gaussian difference-prior model on piecewise-constant and smoothly varying signals.
- Vary signal-to-noise ratio, jump size, and number of jumps.
- Assess both estimation error and credible-interval coverage.
- Explain how banded precision matrices reduce the computational cost of each Gibbs update. Second differences and trend filtering are natural extensions.

Related reading for the empirical Bayes extension is Dongyue Xie (2025), [Empirical Bayes Trend Filtering Through a Variational Inference Framework](#). The goal is to study and compare approaches, without assuming that empirical Bayes trend filtering is a new construction.

7 Latent Autoregressive Models with Cauchy Innovations

Suggested reading: Cima's thesis, Section 2.5, Algorithm 6, and Appendix B.1.

Central problem. Many time series follow a fairly regular dynamical pattern but occasionally experience a large shock or abrupt change. A conventional Gaussian autoregressive model assumes

$$\theta_t = \sum_{j=1}^p \phi_j \theta_{t-j} + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2).$$

Large deviations from the fitted recurrence are therefore assigned very small probability and may substantially distort the estimated dynamics.

Suppose observations are noisy measurements of an underlying latent process:

$$Y_t = \mu + \theta_t + \varepsilon_t.$$

Bayesian approach. Replace the Gaussian process innovations by heavy-tailed innovations:

$$\theta_t = \sum_{j=1}^p \phi_j \theta_{t-j} + \eta_t, \quad \eta_t \sim \text{Cauchy}(0, \tau).$$

Most innovations can remain small, so that the process approximately follows the autoregressive recurrence, while occasional large innovations allow abrupt shocks.

This differs from a Cauchy difference prior in an important way. A first difference prior fixes the recurrence

$$\theta_t \approx \theta_{t-1}.$$

In the autoregressive model, the coefficients

$$\phi_1, \dots, \phi_p$$

are themselves learned from the data, so that the model learns the dynamical recurrence as well as the latent trajectory.

It is also essential to distinguish measurement error

$$Y_t - \mu - \theta_t$$

from the process innovation

$$\theta_t - \sum_{j=1}^p \phi_j \theta_{t-j}.$$

The former represents imperfect measurement of the state; the latter represents a genuine departure of the latent state from its predicted dynamics.

Key questions to study. When do Cauchy innovations improve robustness relative to Gaussian innovations? How does performance depend on shock size and frequency, measurement noise, autoregressive order, and the stability of the fitted dynamics? How accurately can the recurrence coefficients be learned when there are occasional extreme innovations?

Suggested tasks.

- Explain the connection between autoregressive models and Cauchy difference priors.
- Clearly distinguish measurement errors from process innovations.

- Derive and implement the Gibbs sampler for $p = 2$, using proper priors for unknown parameters and initial states.
- Compare a Cauchy-innovation model with a Gaussian-innovation latent AR model and a fixed-order Cauchy difference-prior model.
- Study oscillatory or autoregressive signals with occasional shocks, level changes, or amplitude changes.
- Systematically vary shock magnitude, shock frequency, and measurement noise.
- Examine reconstruction of the latent trajectory, uncertainty in the trajectory, and estimation of the autoregressive coefficients.
- Forecasting may be investigated as an optional extension, with particular attention to stability of the fitted recurrence.

8 Binomial Regression with Cauchy Difference Priors

Suggested reading: Cima’s thesis, Chapter 1, for the difference-prior machinery; additional reading on Pólya–Gamma data augmentation.

Central problem. Suppose that at ordered locations or times we observe

$$Y_i \mid p_i \sim \text{Binomial}(m_i, p_i).$$

For example, Y_i might be the number of purchases among m_i website visitors on day i . Estimating each probability separately by

$$\frac{Y_i}{m_i}$$

ignores the fact that probabilities at neighboring locations are often similar. This can be particularly inefficient when some m_i are small.

At the same time, the underlying probability curve may contain genuine abrupt changes, so an overly smooth model may perform poorly.

Bayesian approach. Work on the log-odds scale

$$\theta_i = \log \frac{p_i}{1 - p_i}$$

and place a heavy-tailed difference prior

$$\theta_{i+1} - \theta_i \stackrel{\text{iid}}{\sim} \text{Cauchy}(0, \tau).$$

The prior shares information across neighboring observations while allowing occasional large changes in the underlying probability curve.

This project extends the Cauchy-difference idea beyond Gaussian sequence regression. There are now two non-Gaussian components: the binomial likelihood and the Cauchy prior. The central computational idea is to use data augmentation to recover a conditionally Gaussian problem.

Pólya–Gamma augmentation converts the binomial logistic likelihood into a conditionally Gaussian form, while the scale-mixture representation of the Cauchy distribution does the same for the

prior. Conditional on the augmentation variables, the update for the entire vector θ has a Gaussian distribution with a banded precision matrix.

Key questions to study. How much does borrowing information across neighboring observations improve estimation when m_i is small? Does the heavy-tailed prior preserve abrupt changes better than a Gaussian difference prior? What happens when $Y_i = 0$ or $Y_i = m_i$? How do unequal sample sizes affect smoothing and posterior uncertainty?

Suggested tasks.

- Specify proper priors on the initial log odds and on τ .
- Explain how the model generalizes Gaussian sequence regression.
- Derive the Pólya–Gamma augmentation of the binomial likelihood and the Gaussian mixture representation of the Cauchy prior.
- Derive the resulting conditionally Gaussian update for the log-odds vector and explain why its precision matrix is tridiagonal.
- Implement the Gibbs sampler, using an existing Pólya–Gamma sampling routine if desired.
- Transform posterior draws of θ_i to estimates and credible intervals for p_i .
- Compare with the separate estimates Y_i/m_i and with a Gaussian difference-prior model.
- Study piecewise-constant and smoothly varying probability curves, unequal positive m_i , and observations for which $Y_i = 0$ or $Y_i = m_i$.

This is an implementation-and-explanation project; no new theoretical result is required. Related work includes Faulkner and Minin (2018), *Locally Adaptive Smoothing with Markov Random Fields and Shrinkage Priors*.

9 Bayesian Finite Population Survey Sampling

Suggested reading: Sudipto Banerjee (2024), Finite Population Survey Sampling: An Unapologetic Bayesian Perspective.

Central problem. Suppose a finite population consists of unknown values

$$Y_1, \dots, Y_N,$$

but we observe only a sampled subset. We may want to estimate the population mean

$$\bar{Y}_N = \frac{1}{N} \sum_{i=1}^N Y_i,$$

the population total, a quantile, or even the entire empirical distribution of the finite population.

Classical survey sampling typically regards the finite population values as fixed. Randomness comes from the sampling design, and estimators such as the Horvitz–Thompson estimator use known inclusion probabilities to obtain design-based guarantees.

Bayesian approach. From a Bayesian perspective, survey sampling can instead be viewed as a missing-data problem. The sampled values are observed coordinates of the finite population vector, while the remaining population values are unknown. A probability model is placed on

$$(Y_1, \dots, Y_N),$$

and the posterior distribution predicts the unobserved coordinates given the sampled ones.

Population means, totals, quantiles, and empirical distributions are then functions of the complete finite population and therefore obtain posterior distributions automatically.

The fundamental conceptual question is:

where does information about the unsampled units come from?

In design-based inference, the sampling design plays the central role. In Bayesian inference, prediction of the missing population values depends on a model relating sampled and unsampled units. The sampling design may be ignorable in some settings and essential in others, particularly when the probability of being sampled depends on the population value.

Key questions to study. When can the sampling design be ignored in Bayesian inference, and when is sampling informative? How do Bayesian estimates relate to classical estimators such as Horvitz–Thompson? What assumptions allow sampled units to inform unsampled units? How sensitive are posterior conclusions to exchangeability, hierarchical structure, spatial dependence, or other modelling assumptions?

It is also useful to go beyond mean estimation. Different models may give similar estimates of the population mean while producing quite different estimates of population quantiles or of the empirical distribution.

Potential directions.

- Explain the Bayesian formulation carefully, identifying the unknown finite population, prior, observed data, likelihood, and sampling mechanism.
- Explain when the sampling mechanism is ignorable and when it must enter the posterior.
- Investigate connections between Bayesian posterior estimates and the Horvitz–Thompson estimator, including models under which the two approaches agree or approximately agree.
- Develop and implement a model from the paper, such as a conjugate Gaussian model, a hierarchical model for multistage sampling, or a spatial model that captures dependence among population units.
- Use simulations or a suitable dataset to compare Bayesian and classical survey estimators for population means or totals.
- Vary the sampling fraction, inclusion probabilities, and strength of dependence among population units.
- Investigate sensitivity to misspecification of the population model and to the sampling design.
- As an extension, study posterior inference for population quantiles or for the complete empirical distribution of $Y_1, \dots, Y_N$.
- Another extension is to compare exchangeable models with models that use covariates, hierarchical structure, or dependence among population units.

The precise scope can be chosen according to the aspects of the paper that are most interesting, but the project should combine exposition of the Bayesian missing-data perspective with a computational investigation of when modelling the unobserved population is helpful and when the conclusions are sensitive to those modelling assumptions.